

K1343 — Muon consistency-check PASSES, and it's a consolidation, not just a check: Elie's "muon = Wallach edge $\nu=3/2$ " and Grace's "rank-1 Korányi-Wolf orbit" are the SAME seat (standard rep theory: the discrete Wallach point $\nu=(r-1)a/2$ IS the representation supported on the rank-1 boundary orbit). And it slots exactly between Lyra's banked seats — tau = Shilov ($\nu \rightarrow 0$), electron = bulk/origin — giving the three charged leptons a single geometric stratification (bulk / rank-1 orbit / Shilov = rank+1 = 3, the F86 support-flag). ONE structure read four ways, NOT four votes. The content still owed: prove mass-ordering = localization-monotonicity. Cal cold-reads.

Keeper (self-directed on "continue," 2026-08-10 ~12:55. I flagged this reconciliation as owed in K1342; I worked it rather than only routing it. It passes, and it consolidates — Elie's new muon result is the missing middle of a picture the corpus already half-had. Tiered honestly; the monotonicity theorem is the real remaining content. Cal cold-reads.)

The reconciliation — PASSES (standard rep theory, not a coincidence)

For a bounded symmetric domain of rank r , multiplicity a , the weighted Bergman spaces H_ν are nonzero exactly on the Wallach set $\{0, a/2, \dots, (r-1)a/2\} \cup ((r-1)a/2, \infty)$. At each discrete point $\nu = k \cdot (a/2)$, the space is supported on the rank- k boundary structure (the relative holomorphic discrete series of a rank- k boundary orbit — Faraut-Koranyi / Arazy). This is the standard bridge between the *analytic* parameter (ν) and the *geometric* stratum (boundary-orbit rank). - D_{IV}^5 : $r=2, a=n_C-2=3 \rightarrow$ discrete points $\nu=0$ ($k=0$) and $\nu=3/2$ ($k=1$), then continuum. - $\nu=3/2$ is the $k=1$ point $\square$ supported on the rank-1 boundary orbit. That IS Grace's rank-1 Korányi-Wolf orbit. So Elie's "Wallach edge $\nu=3/2$ " and Grace's "rank-1 orbit" are the same object in two languages — the analytic edge and the geometric orbit. Not two votes; one seat, two descriptions. Check passes. - ★ Number-caveat (don't conflate the parametrizations): $\nu=3/2$ (Wallach weight), "dim $N_c=3$ " (Grace's orbit dimension), and " $N_\mu \approx 0.55$ " (Lyra's localization amplitude) are THREE different invariants of the SAME orbit. Do not equate $3/2$ with 3. Same seat; distinct labels.

★ The consolidation — the muon is the missing MIDDLE of a picture the corpus already had

Placing Elie's $\nu=3/2$ muon next to Lyra's banked localization seats (F84/F86): the three charged leptons are the **three Korányi-Wolf strata of D_{IV}^5** (open bulk $\square$ rank-1 orbit $\square$ Shilov boundary = rank+1 = 3 strata = 3 generations, the F86 support-flag):

Lepton	Wallach / support	Korányi-Wolf stratum	tripotent rank	Lyra's localization	Physical
electron	bulk / origin (continuum ν , or the center)	open interior	0	$N_e = 1$ (origin)	lightest, most spread
muon	$\nu = 3/2$ (Wallach edge)	rank-1 orbit	1	$N_\mu \approx 0.55$ (Cartan slice)	middle
tau	$\nu \rightarrow 0$ (Hardy / constants limit)	Shilov boundary	2 (maximal)	$N_\tau \rightarrow 0$ (Shilov)	heaviest, most localized

- Four descriptions, one structure: Elie (Wallach edge), Grace (rank-1 orbit), Lyra (F86 strata + localization amplitudes), and the mass hierarchy all land on the same stratification. This is the consistency-web done right — one Korányi-Wolf/Wallach structure of D_{IV}^5 confirmed from four directions, NOT four independent confirmations of a number. (Explicitly guarding my recurring over-determination failure: this adds no evidentiary weight, it *unifies* four readings.)

- Why the muon is the singular one (Elie’s point, now situated): it sits at $\nu=(r-1)a/2 = 3/2 =$ the exact boundary between the discrete series and the continuum — the *limit* of the discrete series. That’s why it has no clean integer-power route (the K1011 null): the analytic structure degenerates precisely at the edge. The electron (bulk) and tau (Shilov, $\nu \rightarrow 0$) sit at *clean* strata; the muon sits on the knife between regimes. A feature, not a gap.

★ The content still owed (this is the theorem, not decoration) — routes #37

The geography above is a strong reconciliation candidate; what makes it a *result* is one monotonicity theorem: - Prove: saturation/localization increases monotonically bulk $\rightarrow$ rank-1 orbit $\rightarrow$ Shilov, and mass tracks it, giving $m_e < m_\mu < m_\tau$ from the stratum alone. The *ordering* is consistent by inspection (heavier = more localized = more boundary); the monotone map from stratum to mass is the charged-lepton module (#37). Until it’s proved, the table is “three leptons = three strata, ordering consistent” — Derived-structure for the muon=rank-1 identification, reconciliation-candidate for the full geography, mass values NOT derived. - Do NOT over-read: this explains the *arrangement and why the muon is hard*, not the mass numbers. State exactly that. (Same discipline as K1342’s muon caveat.)

Tier + routing

- Muon = rank-1 orbit = Wallach edge $\nu=3/2$: Derived-structure (standard Wallach/orbit correspondence). Consistency-check I flagged in K1342: CLOSED, passes.
- The three-strata charged-lepton geography: strong reconciliation candidate, consolidates F84/F86 + Elie 5162 + Grace’s orbit; gives #37 its backbone. Owed: the stratum $\rightarrow$ mass monotonicity theorem (Grace + Elie + Lyra).
- Corpus reconnect (Casey’s standing directive): this is Linear Algebra on D_{IV}^5 — the three generations ARE the three strata of the Wallach set of the weighted Bergman spaces of the one operator; the leptons are three support-classes of one representation family. Ties the master frame (one operator, its Wallach spectrum = the generation ladder).

— Keeper, K1343, 2026-08-10. Muon reconciliation CLOSED (passes): $\nu=3/2$ Wallach edge IS the rank-1 Korányi-Wolf orbit (discrete Wallach point $\nu=(r-1)a/2$ supported on rank-1 orbit; standard). Number-caveat: $\nu=3/2 \neq \dim 3 \neq N_\mu=0.55$, three labels one seat. CONSOLIDATION: the 3 charged leptons = the 3 Korányi-Wolf strata (bulk/origin=electron, rank-1 edge=muon, Shilov $\nu \rightarrow 0$ =tau) = rank+1 = F86 support-flag; matches Lyra’s localization ($N_e=1, N_\mu \approx 0.55, N_\tau \rightarrow 0$) + mass hierarchy. ONE structure four readings, not four votes (guards over-determination). Muon singular because it’s the discrete/continuum edge (explains K1011). OWED = the stratum $\rightarrow$ mass monotonicity theorem (the real content of #37). Cal cold-reads. Nothing pushed.